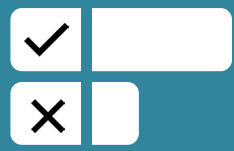


# Protecting Yourself and Your School from Cybercrime

John Greenwood  
Cyber Security Advisor  
Norfolk & Suffolk Constabularies

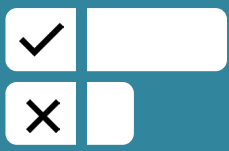
*[john.greenwood@suffolk.police.uk](mailto:john.greenwood@suffolk.police.uk)*



# What is your role?

1. School Governor
2. IT Manager / Technician
3. Teacher
4. Other





# Which of these have you heard of?



National Cyber  
Security Centre

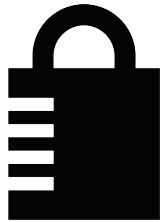


**NCA**  
National Crime Agency



THE  
**EASTERN  
CYBER  
RESILIENCE  
CENTRE**

Cyber  
Aware



**CYBER  
CHOICES**



**POLICE**  
CYBERALARM



**CYBER  
ESSENTIALS**

**ActionFraud**

National Fraud & Cyber Crime Reporting Centre

 [actionfraud.police.uk](https://actionfraud.police.uk) 

# Overview and Aim

Cyber security is growing concern for both organisations and individuals. Cybercrime rose substantially during the COVID 19 pandemic, which saw an increase in remote working arrangements and the use of own devices, much of which continues today.

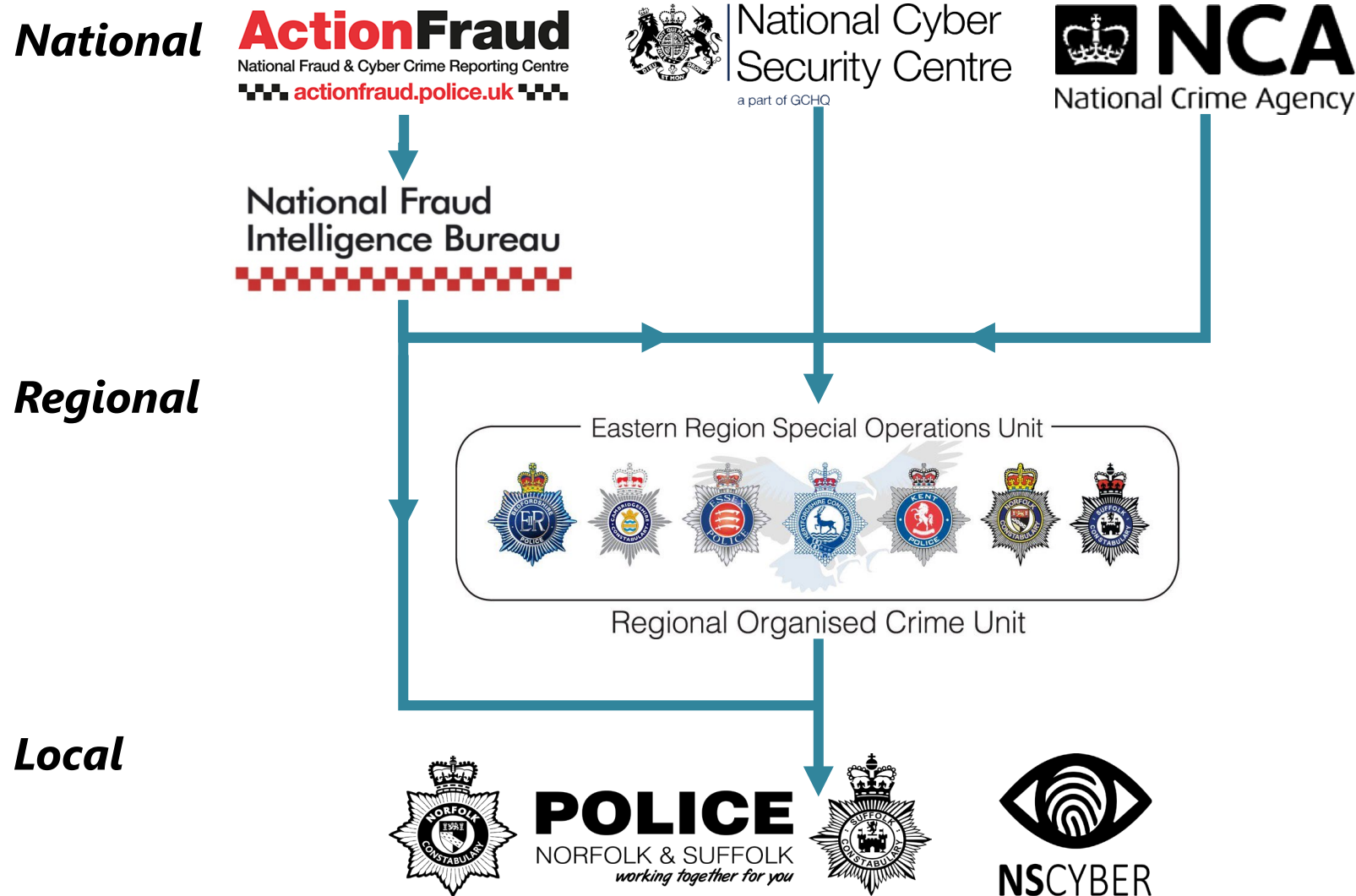
- **Preparing for when, not if**
  - Treating cyber security as a strategic concern
  - Response planning
- **Protecting Yourself and Others from Cybercrime**
  - Methods used by cybercriminals
  - How to become a hard target

# Objectives

**By the end of this session, you will be able to:**

- Recognise the foundation that a cyber security strategy provides to how an organisation responds to a cyber security incident
- Recognise the critical role that social engineering plays in cybercrime
- Minimise your digital footprint
- Create and manage strong, unique passwords
- Recognise the importance of changing a default password
- Apply two-step/multi-step verification to your accounts where available
- Take measures to defend yourself against common cyber threats

# National Cyber Security Strategy



**Cyber *Dependent* Crime**

**Pursue**  
Investigation

**Prevent**  
Stop Involvement



**Protect**  
Increase Defence

**Prepare**  
Increase Resilience

# Hostile Actors & Threats

## **Includes:**

- *Cyber Criminals*
- *Hacktivists*
- *Terrorists*



*Hackers*



*Nation States*



*Insiders*

## **Includes:**

- *Students*
- *Staff*



Phishing



Account  
Compromise



Email  
Compromise



Denial of Service  
(DoS / DDoS)



Malicious  
Software



Data Loss

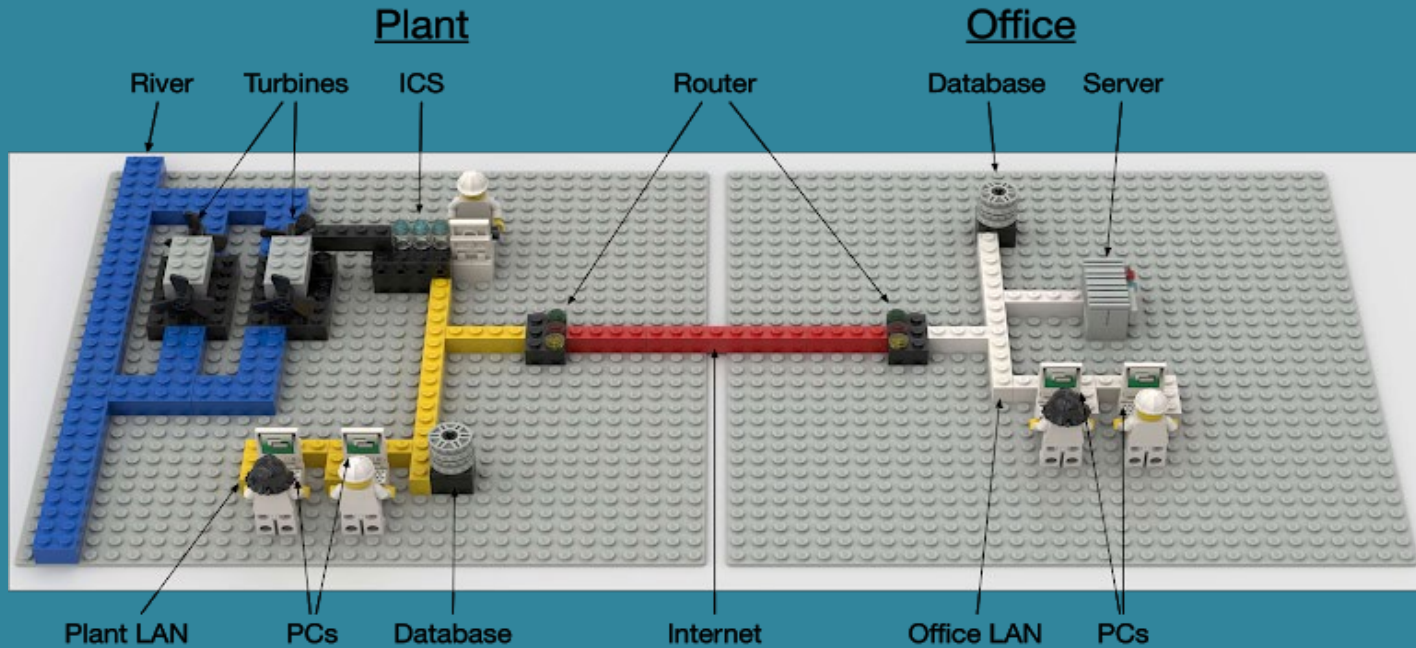
Accidental



Malicious



# Lego Decisions and Disruptions



A tabletop/role-playing game about security in industrial control systems, players are challenged with managing the security of a small utility company.



# Famous Last Words?

***"It won't happen to us, we are too small"***

***"Cyber security is IT's responsibility ..."***

***"Our cyber security is good ..."***

***"Hackers don't target schools ..."***

# Background Question:



**How would your school/college respond to a cyber security incident?**

For instance:

- Account and/or Email Compromise
- Denial of Service attack
- Malware infection

# Cyber Security Breaches Survey 2022 - Education

## **Schools and colleges are targeted by cyber criminals!**

- Identified cyber security breaches:

**Primary Schools**

**41%**

**Secondary Schools**

**70%**

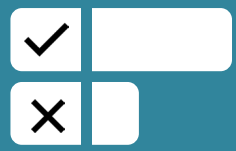
**Further Education Colleges**

**88%**

- Phishing identified as the most common threat (over 88% of identified attacks)

***Key word: “identified” - Actual figure likely to be higher!***

**Over 90% of schools consider cyber security as a priority**



# Does your school have a cyber security strategy and/or a cyber incident response plan?

1. Yes
2. No
3. Don't Know



# What is a Cyber Security Strategy?

A high-level plan for how the organisation, its technological assets and data are protected against accidental or malicious loss/damage.

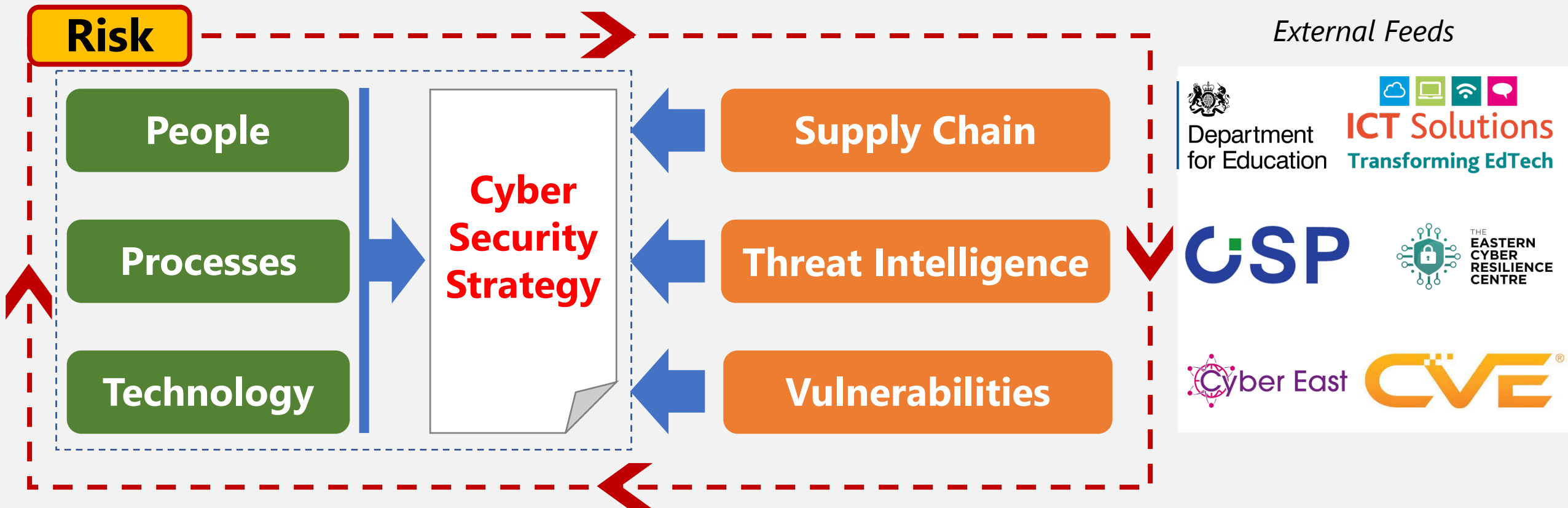
- A 'living document' which is reviewed, updated and tested regularly
- The foundation for incident response plans and playbooks
- Requires the attention of school management and governors
  - Everyone has their part to play though

# Why Have a Cyber Security Strategy?

- It's Proactive!
  - Preventing cybercrime is the best defence against it
  - Schools have unique challenges
- Underpins any response plans the organisation has to:
  - Enable continuity and recovery
  - Disrupt a detected cyber security incident in-progress
  - Contain a breach and limit damage
- Cybercrime is a long term and increasing threat

# Developing a Cyber Security Strategy

- Consider cyber risk from **inside** and **outside** of school/college
- Inform and refine strategy using reputable and reliable external feeds



# Incident Response Plan

***The starting point in responding to a cyber security incident***

## **Should Include:**

- Key contacts
- Escalation criteria and critical decision process
- Process covering incident lifecycle
- At least one conference number
  - Always available for urgent incident calls
- Guidance on legal and regulatory obligations

## **Enhanced by:**

- Simple checklists for ease and convenience
- Methods to document and track an incident
- Technical guidance on:
  - Incident analysis and containment,
  - Remediation and recovery
- Playbooks for specific types of incident

**Derived from official NCSC Incident Management guidance at:**

<https://www.ncsc.gov.uk/collection/incident-management/cyber-incident-response-processes>

# Testing and Refinement

- The cyber landscape is constantly changing.
  - Threats
  - Vulnerabilities
- Strategies, response plans and playbooks need to be tested and refined
  - Ensure they are fit for purpose



**NCSC Exercise in a Box**

<https://www.ncsc.gov.uk/information/exercise-in-a-box>

# Cyber Insurance and/or Risk Protection Arrangement

***Any kind of cyber insurance can be part of a robust cyber security strategy and response plan...***

***...but NOT a substitute!***

- Always check cyber insurance policy details carefully
  - Claiming could be conditional
  - Are there any cover limitations?



<https://www.ncsc.gov.uk/cyberessentials>



<https://cyberalarm.police.uk>



---

PROTECTIVE SERVICES | SPECIALIST CRIME AND INTELLIGENCE





- **Social Engineering**
- **Digital Footprint**
- **Phishing**
- **Account Compromise**
- **Malicious Software**



# Social Engineering

- The deliberate manipulation of a human target
  - Performing specification action(s) e.g.
    - Divulging sensitive information
    - Opening an email attachment
    - Visiting a website
- Humans are the common denominator in cyber security breaches
  - **98%** of Cyber Attacks **rely** on Social Engineering  
(<https://purplesec.us/resources/cyber-security-statistics>)

***“Criminals don’t  
hack in, they log in”***

*~ Detective Inspector  
David Parkin (ret).*

# What is a Digital Footprint?

A term used to describe the entirety of information about you publicly posted online by you or others. This includes audio, photo, text and video content.

## Why should I be concerned?

- Your personal data has value – guard it well!
- Criminals have a myriad of uses for such information:
  - Committing offences such as extortion, fraud, harassment etc.
  - Target research



STAPLETON  
ROAD S.W.

LET & MANAGEMENT  
**D**  
JOHN DEAN  
020 8673 8888

The walker wyatt coffee shop

coffee



# Minimising your Digital Footprint

Be mindful of **what** information is publicly shared online about **you** and **who** can access it

- Consider registering for data breach notifications

'--have i been pwned?

<https://haveibeenpwned.com>

- What does your out of office auto-response reveal about you?
- Do any apps you use need access to things such as contacts and location?

# Digital Footprint and Social Media

What do connections/friends/followers **need** to know?

- Not everyone is who they say they are
- Consider what details are potentially useful to someone with a malicious agenda

## Take a 'privacy-first' approach in the use of social media platforms

- Ensure privacy settings are correctly set to make social media profiles private
- Prevent being "tagged" in a publicly viewable post
- Regularly audit connections/friends/followers
- Be mindful of what information you are sharing on social media platforms

# Phishing - General



Messages from senders disguising themselves as a trustworthy entity that aim to make recipients click a bad link, open an attachment and/or disclose sensitive information

- *Images of text to try and trick filters*
- *Email addresses and web domains with typos are used*
- *Numbers can be spoofed*
- *Sense of urgency or threats*
- *Could contain legitimate links*
- *Links to fake websites*
- *Malicious attachments*
- *The gateway to committing further cyber offences*

# Anatomy of Phishing

## *Broad Attack Vector*

### **Phishing**

Messages are sent  
or calls made en masse

#### ***High Quantity of Attacks***

- *Messages are broad and not personalised*
- *May be followed up with Spear Phishing*

#### ***Often acquired from data:***

- *Available in the public domain*
- *Leaked or stolen from breaches elsewhere*

## *Targeted Attack Vector*

### **Spear Phishing**

Personalised to the  
target(s)

#### ***Research Required***

- *Personalised to a specific target*
- *More believable*

#### ***Additional Vectors:***

- *Business E-Mail Compromise*
- *Potentially followed up with further attacks*

### **Whaling**

Targeted at high-level  
decision makers

# Genuine? or Phishing?

REMITTANCE ADVICE, Payment 0110947

1

AD

ACCOUNTING DEPT <info@johnnywas.com>

Tue 01/02/2022 16:00

To:

Cc:



SUP12142021.html

378 KB



**Payment ID: 9050548**

**Vendor ID : B3KUUNI**

Your payment has now been made, and attached is the payment details with a full payment summary.

Please allow 3 working days from the payment date stated on the remittance advice for the funds to clear your account.

If you have any questions regarding your payment, please direct all enquiries to your local payments team whose contact details are stated on the attached payment advice.

Thank you, your business is important to us!

Name anonymised

# Attempt to Run Malicious Software

~~Genuine~~

Phishing

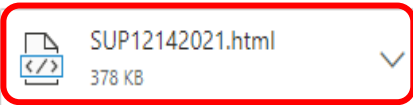
Recipient is targeted

- Social Engineering:**
- Attempting to look “official”
  - Attempting to provoke a sense of urgency and an emotional response

REMITTANCE ADVICE, Payment 0110947

ACCOUNTING DEPT <info@johnnywas.com>

Suspicious email address



Suspicious attachment  
(Name, size and format not atypical of a financial document)  
(possibly contains malicious software)

Payment ID: 9050548

Vendor ID : B3KUUNI

Your payment has now been made, and attached is the payment details with a full payment summary.

Please allow 3 working days from the payment date stated on the remittance advice for the funds to clear your account.

If you have any questions regarding your payment please direct all enquiries to your local payments team whose contact details are stated on the attached payment advice.

Thank you, your business is important to us!

Wants you to open the attachment

Name anonymised

**secure.accountinforming.com** ×

Your connection to this site is private.

Permissions **Connection**

 The identity of this website has been verified by GeoTrust DV SSL CA - G4. No Certificate Transparency information was supplied by the server.  
[Certificate information](#)

 Your connection to secure.accountinforming.com is encrypted using a modern cipher suite.

The connection uses TLS 1.2.

The connection is encrypted and authenticated using AES\_128\_GCM and uses ECDHE\_RSA as the key exchange mechanism.

 **Site information**  
You have never visited this site before today.

[What do these mean?](#)



Fake URL (not legitimate PayPal!)

## Log in to your account

Email address

Password

Log In

[Forgot your email address or password?](#)

Sign Up for Free

All in one pay.

Pick a card, any card, or bank account. It's your money, you choose how to spend it.

Simple. And no hidden costs.

It's free to sign up for a PayPal account, and we don't charge you a transaction fee when you buy something, no matter how you choose to pay.

~~Genuine~~

Phishing

[About Us](#) | [Contact Us](#) | [Fees](#) | [Merchant Services](#) | [Worldwide](#)

[Privacy](#) | [Our Blog](#) | [Jobs](#) | [Legal Agreements](#) | [eBay](#)

Copyright © 1999-2015 PayPal. All rights reserved.  
Consumer advisory- PayPal Pte. Ltd., the holder of PayPal's stored value facility, does not require the approval of the Monetary Authority of Singapore. Users are advised to read the [terms and conditions](#) carefully.

# Individual Phishing Defence

- Always **validate** the sender's authenticity
  - Check the sender's email addresses
  - Hover over links to identify where they go (without clicking on them)
- Be **proactive** in your defence
  - Be mindful of what you share publicly (offline and online)
  - Always question why you're being asked to provide personal data
- Only visit **genuine** organisation websites or call **genuine** contact numbers
  - If in doubt on the legitimacy of any message, do not respond and do not visit any links within
  - Visit the organisation's genuine website or call their genuine number yourself

[www.norfolk.police.uk](http://www.norfolk.police.uk)

Original URL: <http://www.norfolk.police.uk/>. Click or tap if you trust this link.

## Current

Exploit current news, big events or specific times of year (e.g. COVID-19, Fuel, Tax etc.)?

## Authority

Is the message claiming to be from someone official?

## Urgent

Are you told that you have limited time to respond?

## Scarcity

Is the message offering something in short supply or sounding too good to be true?

## Emotion

Does the message make you panic, fearful, hopeful or curious?

## Data

Does the message have basic information about you or request specific data?



# Suspect Phishing? Report It!

If you suspect an email is Phishing, help others by reporting it to:

**report@phishing.gov.uk**

Phishing text messages can be reported by forwarding them to:

**7726**

**As of May 2023:**

Over **20 million** reports have been received, resulting in the removal of more than **129,000** scams across **235,000** URLs.



<https://www.ncsc.gov.uk/collection/phishing-scams>



# Do you re-use passwords?

123456

1q2w3e4r5t

qwerty123

password

blink182

Qwertyuiop

password1

monkey

qwerty

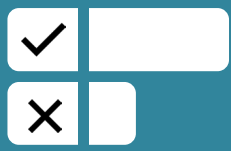
abc123

123456789

p@55word

liverpool





# Which of these passwords is the strongest?

1. **qwerty123**
2. **Pa55w0rd**
3. **EpicCamelMountain35%**
4. **oe4D!3£**

It would take a computer about

**6 minutes**

to crack your password

It would take a computer about

**42 quintillion years**

to crack your password



# Creating a Strong Password

## Think 'passphrase'

1. **Three Random Words** (mixture of upper and lower case)

**e.g. TeaBookMouse**

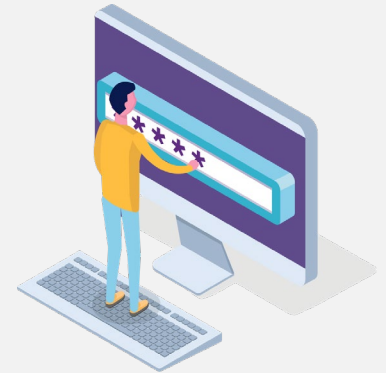
2. Add some complexity by inserting **numbers** and **special characters**

**e.g. TeaBook1Mouse@22**

*Do not substitute letters with characters and numbers*

- Password managers and/or browsers can save your passwords
  - Ensure these are protected just as well!

Ensure passwords are unique, **do not re-use** them



# 2SV Worked Example

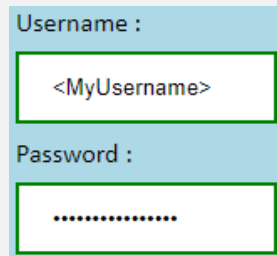
## Scenario:

Logging into an online account which has 2SV enabled

### 2SV Process

1) Something You Know

Password:  
*TeaBook1Mouse@22*



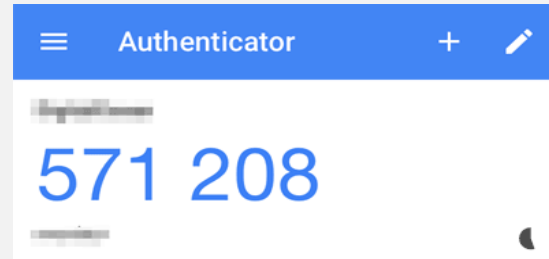
Username :  
<MyUsername>

Password :  
.....

**Correct**

2) Something You Have

Smartphone:  
*Authenticator App*



Authenticator

571 208

Account Access  
Granted

Access granted if  
**both** correct

Account Access  
Denied

Notification of  
access attempt



**Do you use 2SV  
for your personal email and/or social  
media?**



# Account Compromise

- Often the objective of a phishing attack
  - Getting you to unwittingly divulge account username(s) and password(s)
- Business Email Compromise
- Intending to gain unauthorised remote access to data and devices
  - Intent to commit further offence(s) e.g. deploy malicious software such as Ransomware
- Selling of compromised accounts
  - Especially business email and social media accounts

# Business E-Mail Compromise

- Often used in Spear Phishing and/or Whaling
  - Pretending to be someone internal
- Very convincing looking emails, often requesting a payment
  - Requests to change regular payment details (Mandate Fraud)
  - Sudden payment request from a senior figure to a specific account (Boss Fraud)
  - Fake invoices and invoice interception
- Malicious Software Deployment

# Account Compromise - Defence (1)

## Direct Account Security Measures

- Protect accounts using strong, unique passwords with 2SV/MSV
- Never share usernames and passwords with anyone

## Be Vigilant

- Always be willing to double check anything via a different medium
- Check email accounts regularly for any suspicious forwarding rules
- Report suspicious activity immediately

# Account Compromise - Defence (2)

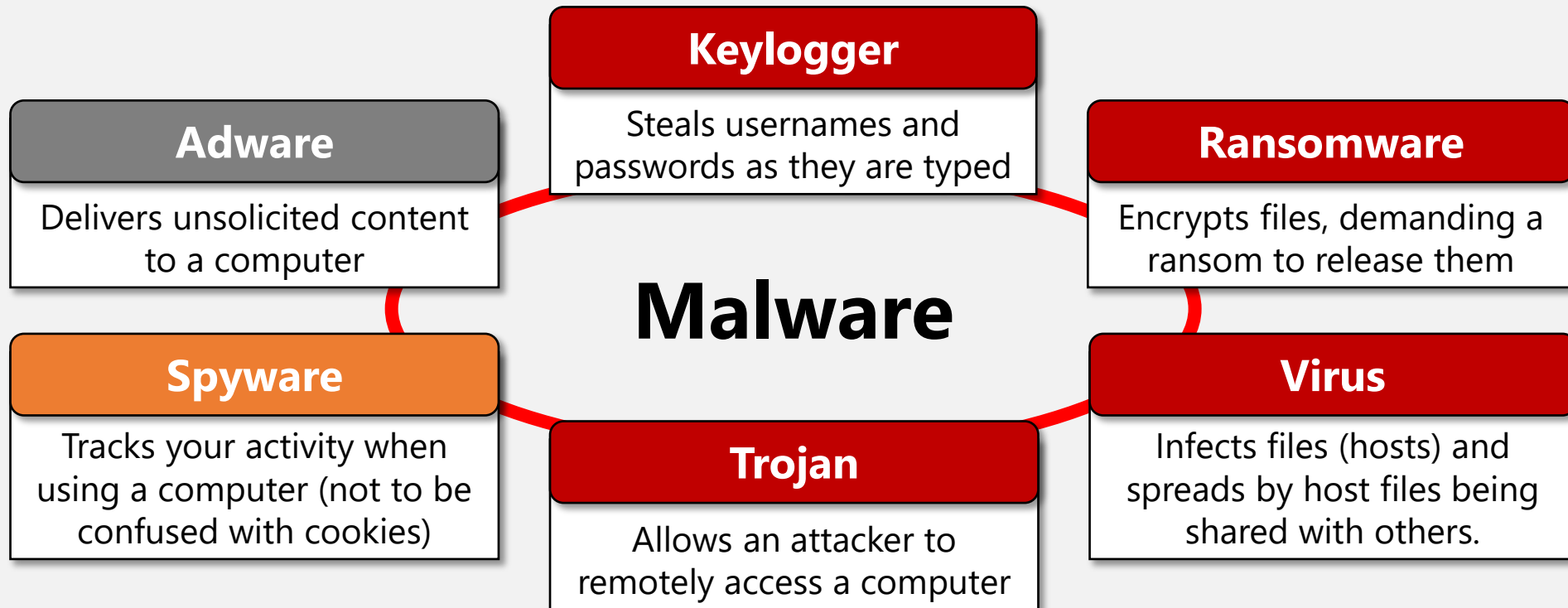
## Be Proactive

- Do not re-use passwords across multiple accounts
- Have recovery options in place for if an account was to become inaccessible
- Be wary of oversharing organisational information
- Always treat requests for payment with suspicion
  - Verify if the request is genuine before doing anything

# Malicious Software

Often simply referred to as “Malware”:

- Software developed intentionally to be harmful to a computer and/or network.



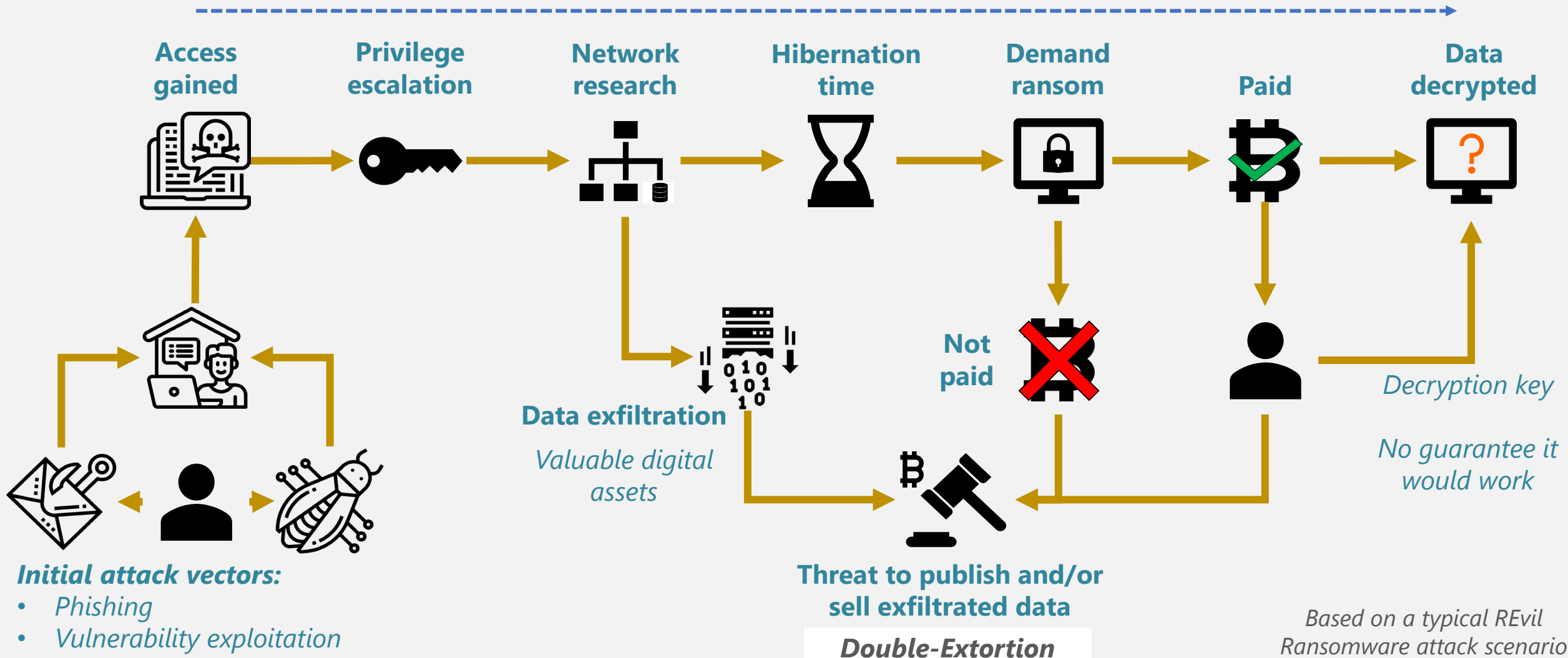
# Ransomware

A type of malware which restricts computer use by locking devices and/or encrypting data, demanding a ransom is paid for its release.

- Organised Cyber Crime
  - Groups of cyber criminals located around the world
  - Organisations are usually the targets
- Ransom typically demanded in a cryptocurrency such as Bitcoin (BTC)
  - May be accompanied by additional threats (Double/Triple-Extortion)
  - Time limit to pay

# Ransomware Attack – Worked Example

*Time – often over weeks or months*



# Impact of Malicious Software

## Organisational

- Operational downtime
- Ineffective data backup could render an attack unrecoverable

## Financial and Legal

- Losing access to operationally critical data resulting in financial loss
- Response and clean up costs
- Consequences and obligations of losing control of commercially sensitive or personal data

# Malicious Software - Defence

## Be Proactive

- Organisational policies and procedures
- Use a real-time monitoring Anti-Virus
  - Schedule regular scanning
- Respond rapidly to infection
- Make regular backups and test them
  - Keep them segregated from the network

## Keep Devices and Software Updated

- Never ignore update notifications
  - Enable automatic updates for your operating system
  - Install manual updates from official sources
  - Anti-Virus software updates and threat definitions

# 'When' not 'If'

Cybercrime is a growing and serious concern for both individuals and organisations

Prevention is always the best form of defence:

- Futile to assume the position of "it will never happen to me"
- Make a plan for '**When**' it happens
- Always keep devices updated and make backups of important/sentimental files
- Make yourself a difficult target

# If You Are a Victim

- Don't stay silent! Report it to ActionFraud!
  - Seek professional assistance if necessary e.g. removing Malware
- Keep copies of / photos of:
  - Logs (server / access / email)
  - Email headers
  - Any related documents
  - Keep any forwarding rules



[actionfraud.police.uk](https://actionfraud.police.uk)



# Summary (1)

- Plan for **WHEN**, not **IF**, a cyber security incident occurs
- Implementing a cyber security strategy is a continuous undertaking
  - Continuously test and refine it
  - Consider the cyber risk from both inside and outside of school/college
  - Requires senior management and governor attention
  - Inform your strategy with information from reliable and reputable sources
    - E.g. CiSP, ECRC, Cyber East, CVE etc.
    - Do not underestimate the value of networking
- A cyber security strategy underpins the response to incidents

# Summary (2)

- Humans are the common denominator in cyber security breaches
  - Social engineering is a critical component of the majority of cyber attacks
  - Minimise your digital footprint, don't make it easy for cyber criminals to target you
- Always use strong and unique passwords
  - Always change default passwords
  - Think 'passphrase'
    - Longer is stronger
    - Three Random Words
- Use 2SV/MSV where available

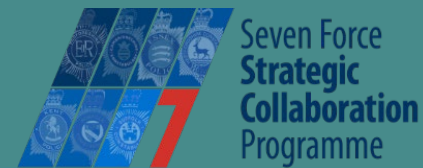
# Summary (3)

- Be proactive in your defence against cyber criminals
  - Keep devices up to date (including anti-virus software)
  - Keep regular, segregated and tested backups
  - Preventing cybercrime is the best defence
- Always respond rapidly if you become a victim of a cybercrime
  - Threat containment and damage limitation
  - Report to Action Fraud



# THE **EASTERN CYBER RESILIENCE CENTRE**

[www.ecrcentre.co.uk](http://www.ecrcentre.co.uk)



Home Office



# What is the ECRC?

**A collaboration between Policing, industry experts and academia to increase cyber resilience in SMEs in the Eastern region.**

- ▶ Raises awareness of the ongoing cyber threat(s) to SMEs, public and third sectors.
- ▶ Offers affordable cyber services to organisations across the Eastern Region to mitigate that risk.
- ▶ Protects and prepares organisations to be more resilient.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <h3>Security Awareness Training</h3> <p>The training is focussed on those with little or no cyber security or technical knowledge and is delivered in small, succinct modules using real world examples.</p> <a href="#">Find Out More</a>                                                                                                                                                                                                                        |  <h3>Corporate Internet Investigation</h3> <p>This service may be used to learn what is being said on the internet about an organisation, what information employees are releasing or if there are any damaging news stories, social media posts or associations.</p> <a href="#">Find Out More</a>                                                                                                             |  <h3>Individual Internet Investigation</h3> <p>The information gathered in this type of investigation might be used to support pre-employment checks, to manage potential threats to a Director of an organisation or their families, or to understand more about a specific person of interest.</p> <a href="#">Find Out More</a>                                                                                                                   |
| <h2>WHAT WE DO</h2>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  <h3>Knowledge &amp; Protection</h3> <ul style="list-style-type: none"><li>• We coordinate cyber risk and protection information between policing and business</li><li>• We help business understand what's relevant to them</li><li>• We help you access NCSC free resources</li></ul> <h4>HOW WE DO IT</h4> <ul style="list-style-type: none"><li>• E- bulletins</li><li>• Seminars &amp; workshops</li><li>• Events &amp; briefings</li><li>• Forums</li></ul> |  <h3>Membership</h3> <ul style="list-style-type: none"><li>• We provide regular e-updates</li><li>• We deliver affordable testing and training services</li><li>• We are a place to find trusted and accredited suppliers</li></ul> <h4>HOW WE DO IT</h4> <ul style="list-style-type: none"><li>• Membership benefits</li><li>• Affordable services</li><li>• Trusted Partners (Certified providers)</li></ul> |  <h3>Skills &amp; Talent</h3> <ul style="list-style-type: none"><li>• We provide real world experience for emerging university cyber talent</li><li>• We develop real world commercial skills for students</li><li>• We provide transferrable skills opportunity for veterans</li></ul> <h4>HOW WE DO IT</h4> <ul style="list-style-type: none"><li>• Student mentoring</li><li>• Student paid work experience</li><li>• Student training</li></ul> |
|  <h3>Security Policy Review</h3> <p>This service offers a review of your current security policy, how it is written and how it is implemented.</p> <a href="#">Find Out More</a>                                                                                                                                                                                                                                                                                |  <h3>Cyber Business Continuity Review</h3> <p>This service offers a review of your business continuity planning and the resilience of your organisation to cyber-attacks such as ransomware or when attackers take control of your core systems.</p> <a href="#">Find Out More</a>                                                                                                                           | <h3>Partner Resource Support</h3> <p>Student resource will be used to fill temporary resource gaps, support extended resource requirements to support projects, or during incident response.</p> <a href="#">Find Out More</a>                                                                                                                                                                                                                                                                                                         |



# How can the CRCs help you, your staff, customers and your supply chain?

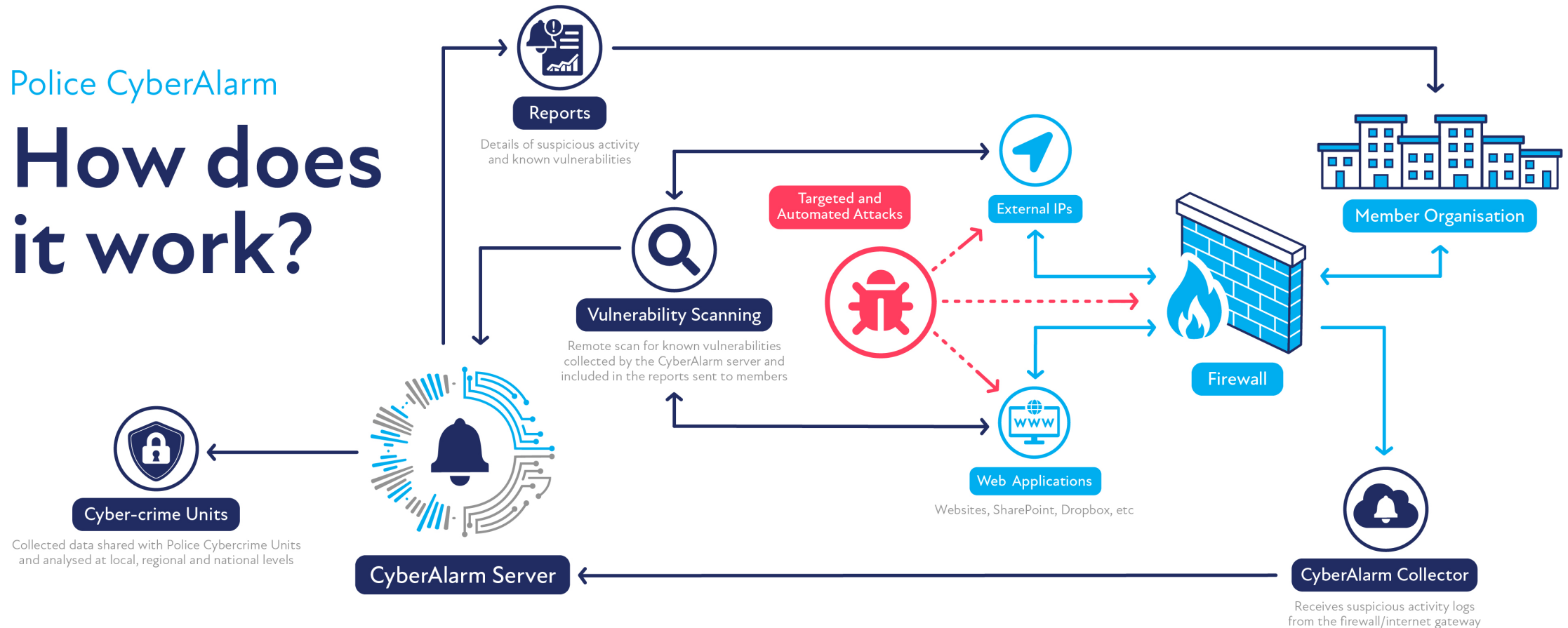
**Improved cyber resilience means less chance of becoming a victim of cyber crime and fraud improved supply chain vulnerability for your business and the wider business community.**

- ▶ Free membership of the ECRC community.
- ▶ Newsletter with regular threat assessment.
- ▶ Webinars and blogs - sector specific content.
- ▶ Signposting to NCSC tools and services.
- ▶ Little Steps Program for Cyber Essentials.
- ▶ Access to Student Services and Trusted Partners.
- ▶ Police Cyber Alarm and NCSC Early Warning.



Police CyberAlarm

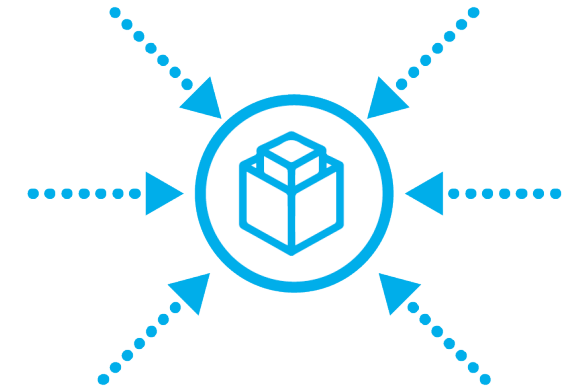
# How does it work?



# What data does Police CyberAlarm collect?

## Police CyberAlarm only collects Firewall 'Metadata'

- Logs collected from internet facing devices
- How data was sent/received this includes:
  - Where the data came from - IP Address
  - Amount of data transferred
  - Port used to transfer the data
- The content of the traffic is not collected
- Police CyberAlarm does not interfere with network operations



### Sample Firewall Log

```
date=2020-01-07
time=13:25:27 timezone="GMT" device_name="CR500ia" device_id=
C070123456-ABCDEF
log_id=010102600002 log_type="Firewall" log_component="Firewall
Rule" log_subtype="Denied" status="Deny"
priority=Information fw_rule_id=3 ips_policy_id=0 appfilter_policy_id=0
in_interface="PortG.16" out_interface="PortB" src_mac=00:0d:48:0a:
05:45
src_ip=172.16.16.95 src_country_code="GB" dst_ip=192.168.5.2
protocol="UDP" src_port=42288 dst_port=53 sent_pkts=0 rcv_pkts=0
sent_bytes=0 rcv_bytes=0
```

## Vulnerability Report Summary

This report details known potential vulnerabilities seen on the target URLs and IP addresses with the CyberAlarm Vulnerability Scanner. For further information and advice on what to do with the details contained in this report, please contact your systems administrator.

| Vulnerability Summary for <a href="https://www.sampleorg.co.uk">https://www.sampleorg.co.uk</a> |                     |
|-------------------------------------------------------------------------------------------------|---------------------|
| Scan Date                                                                                       | 09/06/2020 at 18:03 |
| <b>Total Vulnerabilities Found</b>                                                              | <b>37</b>           |
| <b>High Risk Vulnerabilities Found</b>                                                          | <b>7</b>            |
| <b>Medium Risk Vulnerabilities Found</b>                                                        | <b>12</b>           |
| <b>Low Risk Vulnerabilities Found</b>                                                           | <b>18</b>           |
| Additional Information                                                                          | 79                  |

## Vulnerability Scanning

Cyber Alarm is able to scan external IP addresses. Vulnerabilities found are reported with a Common Vulnerabilities & Exposures (CVE) code.

| Name                       | Type                   | Severity | CVE Code         | Details                                                                                                                                        |
|----------------------------|------------------------|----------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| SQL Injection              | SQL                    | High     | CVE-2019-1010259 | Vulnerable version of SaltStack (Salt 2018.3) discovered                                                                                       |
| Slow HTTP POST             | Denial of Service      | Medium   | CVE-2019-9518    | Some HTTP/2 implementations are vulnerable to a flood of empty frames, potentially leading to a denial of service.                             |
| HTML Injection             | Web Application        | Low      | CVE-2019-0216    | A malicious admin user could edit the state of objects in the Airflow metadata database to execute arbitrary javascript on certain page views. |
| CGI Scanning Consolidation | Information Disclosure | Info     |                  | This service seems to be able to host PHP scripts. This service seems to be able to host ASP scripts.                                          |

## Testimonials

“Cyber Alarm was implemented with ease and provides essential proactive intelligence of external cyber security threats. With the regular reporting received from Cyber Alarm, we are now equipped to secure against any new vulnerabilities as they are detected.”

Gordon Perkins LGSS at Northamptonshire County Council

“The report we receive from the Cyber Alarm has been invaluable, not only are we able to block IP's attempting malicious attacks, it also helps inform our Board of Directors of the scale of the problem in a non-technical way. This ensures that resources within the business can be directed proportionately and appropriately.”

Nicola Petts Group IT IG Vendor Manager at Christchurch Group

# Aims of the CYBER CHOICES Programme

Explaining the difference between legal and illegal cyber activity

Encouraging individuals to make informed choices in their use of technology

Increasing awareness of the Computer Misuse Act 1990

Promoting positive, legal cyber opportunities

[www.cyberchoices.uk](http://www.cyberchoices.uk)



- To deter individuals from getting involved in cyber dependent crime, and prevent re-offending
- To promote legal and ethical use of skills, including opportunities in cybersecurity

 *Average age of a Cyber Criminal is **17.5***

 *Common pathway into Cyber Crime is.....*  
*gaming → mods → hacking*



***31% of local teens anonymously admitted to trying to compromise someone else's account without permission (social media or similar)***

***63% said they have been kicked off a game***

458 Students asked (Yr10- Post 16) in 2022/23



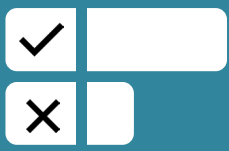
PROTECTIVE SERVICES | SPECIALIST CRIME AND INTELLIGENCE



# Summary of CYBER CHOICES

- Cyber Choices is a National Crime Agency programme aimed at young people
  - Engage and Educate
  - Identify, Intervene and Inspire
- Financial gain often not the primary motivator to commit cybercrime
- A 'prevent duty' of sorts for cyber dependant crime
  - Referenced in Keeping Children Safe In Education (Annex B pg 144)

[www.cyberchoices.uk](http://www.cyberchoices.uk)



# Thank You!

**As a result of this training, do you feel more informed on the cyber security for both yourself and your school?**





National Cyber  
Security Centre

a part of GCHQ

[ncsc.gov.uk](https://ncsc.gov.uk)

Cyber  
Aware   
[cyberaware.gov.uk](https://cyberaware.gov.uk)

**ActionFraud**  
National Fraud & Cyber Crime Reporting Centre  
 [actionfraud.police.uk](https://actionfraud.police.uk)   
[actionfraud.police.uk](https://actionfraud.police.uk)



[2fa.directory](https://2fa.directory)

'--have i been pwned?

[haveibeenpwned.com](https://haveibeenpwned.com)



[getsafeonline.org](https://getsafeonline.org)



TO STOP FRAUD™

[takefive-stopfraud.org.uk](https://takefive-stopfraud.org.uk)



[friendsagainstscams.org.uk](https://friendsagainstscams.org.uk)



[nationaltradingstandards.uk](https://nationaltradingstandards.uk)



ARE YOU SCAM-SAVVY?

[quiz.takefive-stopfraud.org.uk](https://quiz.takefive-stopfraud.org.uk)



Jigsaw | Google

[phishingquiz.withgoogle.com](https://phishingquiz.withgoogle.com)



**CYBER  
CHOICES**

[cyberchoices.uk](https://cyberchoices.uk)

# Additional Resources:

- **NCSC 10 Steps to Cyber Security**  
<https://www.ncsc.gov.uk/collection/10-steps>
- **Cyber Security Information Sharing Partnership (CiSP)**  
<https://www.ncsc.gov.uk/section/keep-up-to-date/cisp>
- **Eastern Cyber Resilience Centre**  
<https://www.ecrcentre.co.uk>
- **Cyber East**  
<https://www.cybereast.co.uk>
- **Local Police Forces**  
<https://www.norfolk.police.uk/advice/cybercrime>  
<https://www.suffolk.police.uk/advice/cybercrime>
- **NCSC Active Cyber Defence**  
<https://www.ncsc.gov.uk/section/active-cyber-defence/introduction>
- **NCSC Cyber Aware**  
<https://www.ncsc.gov.uk/cyberaware/home>
- **NCSC Exercise in a Box**  
<https://www.ncsc.gov.uk/information/exercise-in-a-box>
- **Police CyberAlarm**  
<https://www.cyberalarm.police.uk>
- **CVE**  
<https://www.cve.org>



**CyberProtect@Suffolk.police.uk**  
**CyberProtect@Norfolk.police.uk**



**@NSCyberCrime**



**[norfolk.police.uk/advice/cybercrime](https://norfolk.police.uk/advice/cybercrime)**  
**[suffolk.police.uk/advice/cybercrime](https://suffolk.police.uk/advice/cybercrime)**



**[NSCyber.com/BusinessFeedback](https://NSCyber.com/BusinessFeedback)**

**Online Schools Staff Training  
(throughout the year)**

